

Geomec 5.4 Release Notes

Geomec 5.4 has the following new features and improvements:

1) Licensing:

- Windows 10 compatible
- Improved borrowing dialog now shows the borrowed OIV licenses
- Documentation for borrowing and license troubleshooting has been improved
- Before calculation Geomec checks for a Diana license
- During calculation OIV is deactivated

Note: OIV can still show a "license lost" dialog; as long as you don't press OK, this should not interfere with the calculation

2) Default GM5 File Format:

- HDF5 file format, accessible with external tools
- Smaller memory footprint due to delayed result loading
- Backwards compatible with GM4 (using Save As)

3) Wellpaths:

- Unique wellpath input definitions in the Data Storage tree
- Automatic copies of the wellpath input definition to main and applicable child models [OPTIONAL]
- Detection of unrepresented formations due to mesh size
- Cylindrical hull for distributed property mapping
- Techlog LAS 3.0 export
- Automatic detection of import file type (Ascii / Petrel), and other improvements to import
- Small improvements, like flipping Depth, azimuth/inclination/AHD available as properties in visualization and export

4) Calculation:

- Diana 10.1 release version is included
- Quadratic elements can be defined per formation
- Mixture (Containment) calculation has been corrected
- Depleting formations based on calculation type: temperature change for heat flow, pressure change for all others
- Stress initialization in D0

5) General Import/Export:

- Handling FIELD/SI units has been corrected
- Result macro can be imported and exported as XML libraries

6) Model Consistency Guard:

- Quick visual feedback in the calculation toolbar shows whether model is ready to be calculated
- Overview and settings dialog behind the quick visual feedback
- Consistency settings:
 - Automatically clear results of child models if parent model changes

- Automatically copy wellpaths to applicable models (intersecting with mesh)
- Automatically try alternatives for pointset dimension if the convex hull cannot be created
- Automatically export the active result macros

7) Visualization:

- The old deprecated Volumetric Strain has been removed, and existing values are automatically converted to the new Volumetric Strain under the Strains branch
- User defined properties can be visualized on the Material Parameters branch
- Reinstated creation of horizontal/vertical cross sections

8) Settings:

- Parallel optimizations for visualization and Diana are now turned on by default
- Multiple temporary folders can be set

9) Help:

- Updates for changed functionality
- Help file and additional documents are available in Geomec's Start Menu folder

10) External programs and add-ons:

- CORA user interface updated to be compatible with Windows 10
- GVT experimental version is included:
 - Based on Neil Hodgson's python version, extended to hexahedron models
 - Takes Vp from the material models
 - Import/Export of velocity files is not yet supported

11) Smaller issues:

- Qt libraries are upgraded to version 5.6.1
- Only one of lateral or normal strain needs to be defined
- Compaction capacity is fixed
- Analysis point rendering is fixed

Known issues in 5.4:

- 1) GVT is an alpha test version. It doesn't support export/import of velocity files.
- 2) Qt 5.6.1 has problems with handling symbol characters. As a workaround, "°C" is currently rendered as "C", but there may be more issues.
- 3) The option to show convex hulls doesn't include cylindrical hulls yet.

Known issues from 5.3 still present in 5.4:

- 1) Well casing zoom-in model is not yet properly working
- 2) Contour lines are absent in the legend, and can only be shown for the property as the shading shows
- 3) Using graph ranges in the 2D wellpath view may clip the wellpath on the right, depending on Geomec's window size